

APELLIDO DEL ALUMNO: NOMBRE:

CORRIGIÓ: REVISÓ:

T1	T2	P1	P2	P3	P4	CALIFICACIÓN

Todas las respuestas deben ser justificadas adecuadamente para ser tenidas en cuenta.

No resolver el examen en lápiz. Duración del examen: 2 horas

Condición de aprobación (6 puntos): tres ejercicios correctamente resueltos (uno de T1 o T2 y dos de P1, P2, P3 o P4).

T1) a) Defina superficie parametrizada y punto regular de una superficie.

b) Sea R_0 la recta normal en $\bar{A} = (2, 1, 3)$ a la superficie S de ecuación $\bar{X} = (2u^2, v - u, v + u)$ con $(u, v) \in \mathbb{R}^2$; analice si R_0 corta al plano $x + y = z + 5$; en caso afirmativo halle el punto de corte.

T2) a) Grafique el conjunto de integración de la siguiente integral doble:

$$\int_0^{\frac{\pi}{4}} \left(\int_0^{2 \operatorname{sen} \theta} \rho^2 \operatorname{sen} \theta d\rho \right) d\theta$$

b) Expresa dicha integral en coordenadas cartesianas

U7 P1- Dada $T(x, y) = x^2 y + y^2 + x^2$; analice y clasifique extremos locales.

U8 P2 - Sea $\vec{f}(x; y; z)$ solenoidal (Divergencia nula) / $\vec{f}(x; y; 0) = (x, y - 2, y^2)$, calcule el flujo de $\vec{f}$ a través de Σ de ecuación $z = \sqrt{4 - x^2 - y^2}$, indique el versor normal considerado.

U1 P3 - Obtenga la curva ortogonal a la familia dada por $xy = K$, que pase por $(5, 3)$ y luego, elabore una parametrización para dicha curva, para x e y positivos.

U9 P4- Expresa la masa del cuerpo definido por: $z \geq \sqrt{x^2 + y^2}$, $x^2 + y^2 \leq 4$, $z \leq 4$, si la densidad en cada punto es proporcional a la distancia del punto al plano xy .

11/04/2025

T1)

$$\vec{r}_0 = x_0 + \lambda \vec{r}'_0$$

normal

$$\vec{i} \quad \vec{j} \quad \vec{k}$$

$$\vec{x}' = (4\mu, -1, 1)$$

$$\vec{x}_0 = (0, 1, 1)$$

$$\vec{u} \times \vec{r}' = \begin{vmatrix} 4\mu & -1 & 1 \\ 0 & 1 & 1 \end{vmatrix} = (-1-1)\vec{i} + (4\mu)\vec{j} + (4\mu)\vec{k}$$

$$= (-2, -4\mu, 4\mu)$$

$$2 = 2\mu^2$$

$$\mu = \pm 1$$

$$|\mu| = 1$$

$$\sqrt{-\mu} = 1$$

$$v + u = 3 \quad 1 + u + u = 3 \rightarrow 2u = 2 \rightarrow \mu = 1$$

$$v - u = 1 \rightarrow v = 2$$

$$v = 2$$

$$\vec{T} \times \vec{r}' = (-2, -4, 4)$$

$$(1, 2)$$

$$\vec{r}_0 = (2, 1, 3) + \lambda (-2, -4, 4)$$

$$2 + \lambda(-2) = x = 2 - 2\lambda$$

$$1 - 2\lambda = y = 1 - 4\lambda$$

$$3 - 2\lambda = z = 3 + 4\lambda$$

locus?

$$2 - 2\lambda + 1 - 4\lambda = 3 + 4\lambda + 5$$

$$2 + 1 - 3 - 5 = 4\lambda + 2\lambda + 4\lambda$$

$$-5 = 10\lambda \rightarrow \lambda = -\frac{5}{10}$$

$$\exists t \rightarrow \exists \lambda \text{ w.t.}$$

T2) a)

$$\int_0^{\pi/4} d\theta \int_0^{2\sin\theta} p^2 \sin\theta dp$$

$$p \cdot \frac{p^2 \sin\theta}{3}$$

$$x = p \cos\theta$$

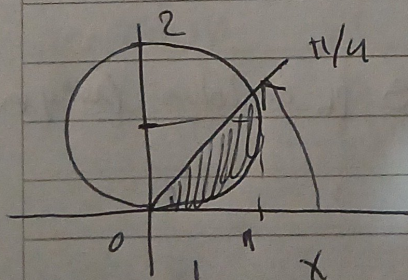
$$y = p \sin\theta$$

$$0 \leq p \leq 2\sin\theta$$

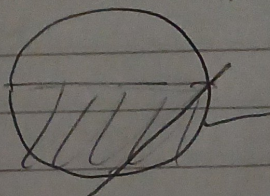
$$p^2 \leq 2p \sin\theta$$

$$x^2 + y^2 < 2y$$

$$x^2 + (y-1)^2 \leq 1$$



$$b) \int_0^1 dx \int_{1-\sqrt{1-x^2}}^1 y dy$$



P1) $f(x,y) = x^2y + y^2 + x^2$ compute & find extremes, admit plane graph

$$\underline{CN} = \nabla f(0,0) = 0$$

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$$(1) f'_x = 2xy + 2x = 0 \rightarrow 2x(y+1) = 0 \rightarrow 2x = 0 \rightarrow x = 0$$

$$(1) f'_y = x^2 + 2y = 0 \Rightarrow x^2 = -2y \quad y+1=0 \rightarrow y=-1$$

$$x=0 \text{ in (1)} \quad 2y=0 \rightarrow y=0$$

$$(0, -1), (0, 0)$$

$$y=-1 \text{ in (1)} \rightarrow x^2 - 2 = 0 \Rightarrow x = \pm\sqrt{2}$$

$$x = \pm\sqrt{2}$$

$$\begin{pmatrix} 0, 0 \\ (-\sqrt{2}, -1) \\ (+\sqrt{2}, -1) \end{pmatrix}$$

$$f''_{xx} = 2y + 2$$

$$f''_{xy} = 2x$$

$$f''_{yx} = 2x$$

$$f''_{yy} = 2$$

$$H_{(x,y)} = \begin{vmatrix} 2y+2 & 2x \\ 2x & 2 \end{vmatrix}$$

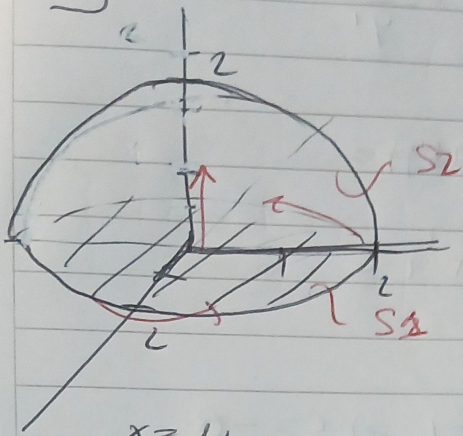
$$H(0,0) = \begin{vmatrix} 2 & 0 \\ 0 & 2 \end{vmatrix} = \det H > 0 \quad 2 > 0 \rightarrow \text{lok min at } (0,0, f(0,0))$$

$$H(\pm\sqrt{2}, -1) = \begin{vmatrix} 0 & \pm\sqrt{2} \\ \pm\sqrt{2} & 2 \end{vmatrix} = \det = 0 < -8 \text{ It's saddle (not an ext.)}$$

$$H(\pm\sqrt{2}, -1) = \begin{vmatrix} 0 & 2\sqrt{2} \\ 2\sqrt{2} & 2 \end{vmatrix} = \det = 0 < -8 \text{ It's saddle (not an ext.)}$$

Alexandre Nogueira

PG



$$x = u$$

$$y = v$$

$$z = 0$$

$$T(u, v) = (u, v, 0)$$

$$z = \sqrt{4 - x^2 - y^2} \Rightarrow 4 - x^2 - y^2 = z^2$$

$$S = S_1 \cup S_2$$

$$\iint_{S_1} f_m d\sigma + \iint_{S_2} f_m d\sigma = \iiint_V dV \rho = 0$$

$$\log' = 0$$

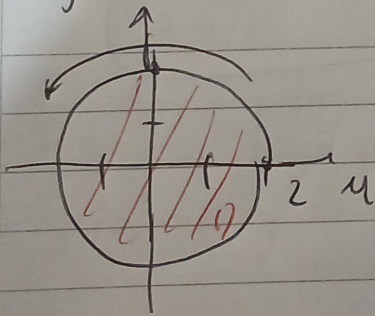
$$T'u = (1, 0, 0)$$

$$T'v = (0, 1, 0)$$

$$T'u \times T'v = \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = (0, 0, 1)$$

$$\iint f(T(u, v)) \frac{|T'u \times T'v|}{|T'u \times T'v|} d\sigma$$

$$= \iint f(u, v, 0) (0, 0, 1) d\sigma = \iint f(u, v, 0) d\sigma$$

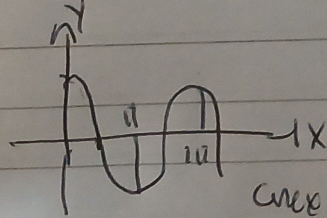


$$\int_0^{2\pi} \int_0^2 r^2 \sin \varphi \cdot r dr d\varphi = \sin \varphi \frac{r^4}{4} \Big|_0^2 = \sin \varphi \cdot 4$$

$$(-1)(4) - (-1)(4) = 0$$

$$u = r \cos \varphi$$

$$v = r \sin \varphi$$



$$0 \leq \varphi \leq 2\pi$$

$$0 \leq r \leq 2$$

$$\left[\frac{\varphi}{2} + \frac{\sin \varphi}{4} \right]_0^{2\pi} = 0$$

$$T(\varphi) = (\cos \varphi, \sin \varphi) = \begin{pmatrix} \cos \varphi \\ \sin \varphi \end{pmatrix}$$

$$T(0) = (1, 0)$$

$$T(\pi/2) = (0, 1)$$

$$\iint f_m d\sigma$$

$$x^{-1} = x^{-1} = \frac{1}{x^2} (-1)$$

P3/ $y = \frac{u}{x} \Rightarrow y' = \frac{u}{x^2} = -\frac{xy}{x^2}$

$$xy = u$$

$$y' = -\frac{y}{x} \rightarrow y' = -\frac{1}{y}$$

$$-\frac{1}{y'} = -\frac{y}{x} \rightarrow dx = -y dy \rightarrow \int dx = \int -y dy =$$

$$\left(\frac{y^2}{2} = \frac{x^2}{2} + c \right) \rightarrow y^2 = x^2 + c$$

$$\frac{25}{2} = \frac{9}{2} = \frac{25}{2} + c = 1 \Rightarrow c = -8$$

SP $\left(\frac{y^2}{2} = \frac{x^2}{2} + 8 \right) \rightarrow y^2 = x^2 + 16$

$$\sqrt{x^2} = \sqrt{y^2 + 16}$$

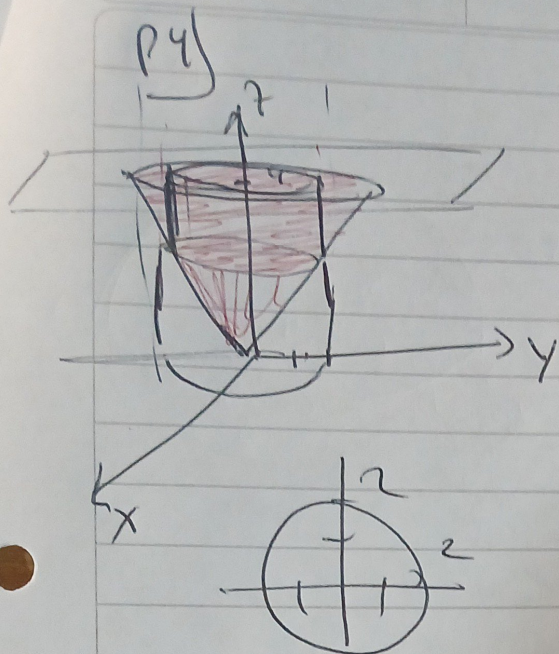
$$x = \sqrt{y^2 + 16}$$

$$|x| = \sqrt{y^2 + 16}$$

$$x = \sqrt{t^2 + 16}$$

$$y = t$$

$$r(t) = (\sqrt{t^2 + 16}, t) \quad t \geq 0$$



$$\rho = u|z|$$

$$\iiint u z \, dV$$

$$u = \iiint = \int_0^2 dr \int_0^{2\pi} d\varphi \int_r^4 dz \cdot r$$

Jacobiano
1

$$z = 4 - r \cos \varphi = 1 \text{ cm}$$

$$\varphi = r \sin \varphi \quad z = z$$

$$= \left. \frac{r z^2}{2} \right|_r^4 = (4r - r^2) \, d\varphi$$

$$z = \sqrt{4r^2 \cos^2 \varphi + 1 \cos^2 \varphi}$$

$$= \sqrt{4r^2 \cos^2 \varphi + \cos^2 \varphi}$$

$$= 4r\varphi - r^2\varphi \Big|_0^{2\pi} = (8r - 2\pi r^2) \, dr$$

$$\frac{8r^2}{2} - \frac{2\pi r^3}{3} \Big|_0^2 = 16 - \frac{16\pi}{3}$$

$$z = \sqrt{4\pi^2}$$

$$z = r$$

hay en eso la
corta r' hay
pero el resultado es

$$28\pi u$$

$$\iiint u z \, dV = 28\pi u$$